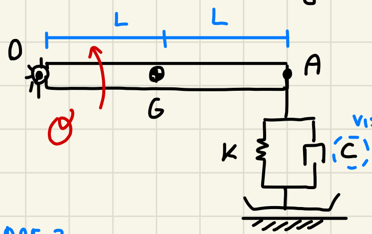
ESE 2

previously we see yet linear systems... ↴ NOW we deal with NOT kinematically linear system

es1

{ bar with uniform mass distribution }



2L total lenght

viscous behav of dashpod

• check this is 1 DOF system... ?

• WRITE the eq of motion of the syst. ?

DOF ? ⇒

1 Riged body: (before any constraint) 3DOF

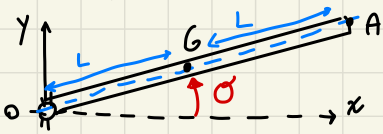
hinge in O −2 DOC (only constr acting on body)

⇓

we can confirm this ← 1 DOF

(θ) ⇐ by chosing a single variable Independent, and if we freeze the indipedent cohord is all frozen

redrawing the system in a generic position (after θ Rotation)



reference system (x, y)

θ angle between a line moving with system and x axis

↓absolute Bar Rotation

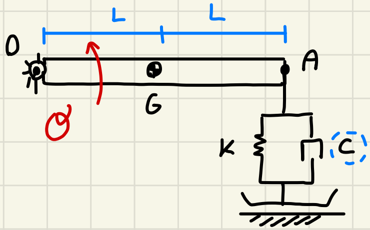
IT is obvious that IF θ = constant ⇒ ALL FROZEN!

any point G, A, O.. is all fixed and defined by θ

$OG = L$
$GA = L$ always same distance (Rigid body)

θ indip cohord to describe motion

⇒ LAGRANGE EQUATION for eq of motion



ϑ JUST scalar Rotation (1 DOF)

We need JUST scalar equation

$$\frac{d}{dt}\left(\frac{\partial T}{\partial \dot\vartheta}\right) - \frac{\partial T}{\partial \vartheta} + \frac{\partial D}{\partial \dot\vartheta} + \frac{\partial V}{\partial \vartheta} = \frac{\delta W_{NC}}{\delta \vartheta}$$

virtual work of NON dissipative, conservative -- forces

express explicitly energy quantity

infinitesimal virtual Rotation displacement

(to obtain LINEARIZED eq. of motion) $\begin{cases} \text{STANDARD Way (1)} \\ \text{shortcut (2)} \end{cases}$

- approach (1) STANDARD

consider any possible large motion → NON Linear eq of motion → Linearize this with series expansion respect $\vartheta, \dot\vartheta, \ddot\vartheta$ (Troncated to 1st order term)

small perturbat.

{(2) use particular result of linearization}

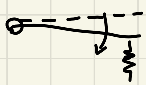
ASSUME

$\vartheta_0 = 0$ equilibrium with HORIZ. BAR

there is some gravitational forces that in STATIC position is balanced by SPRING (elastic) force

- and if we SET the system with undeformed spring at equilibrium horizontal

Instead we expect STATIC position a bit inclined down to generate enough deformation balancing the weight



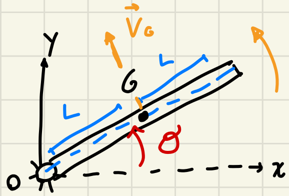
OR use device to adJUST position

So equilibrium @ $\vartheta_0 = 0$ with $\Delta l_0 = 0$

⇒ as long as small $\vartheta < 0$ inclination to balance could be very small for STIFF K → and $y_A \simeq [mm]$ respect $2L \simeq [m]$

infinitesimal RATIO respect $\vartheta = 0$ and ϑ_0 of decreasing

↓ WE can set $\vartheta_0 = 0$



$\dot{\theta}$ angular speed

$$\begin{cases} x_G = L\cos\theta \\ y_G = L\sin\theta \end{cases} \Rightarrow \begin{cases} \dot{x}_G = -L\sin\theta\,\dot{\theta} \\ \dot{y}_G = L\cos\theta\,\dot{\theta} \end{cases}$$

scalar speed components of $\vec{V}_G$

$\vec{V}_G$ for Kinematic theorem
Is perpendicular to OG with magnitude

$$|\vec{V}_G| = L\cdot\dot{\theta}$$
$(\overline{OG}\cdot\omega_{BODY})$

sign convention: $\dot{\theta}>0$ when counterclockwise

$\vec{V}_G$ directed as fingers

• Kinetic energy (only 1 body)

m: body mass
J: inertia of bar respect G

$$T = \frac{1}{2}\left(m\dot{x}_G^2 + m\dot{y}_G^2 + J\dot{\theta}^2\right) =$$

RIGID Body mooving in a plane KÖNIGS Th.

$$= \frac{1}{2}\left(mL^2\sin^2\theta + mL^2\cos^2\theta + J\right)\dot{\theta}^2 = \frac{1}{2}\underbrace{\left(mL^2 + J\right)}_{\bar{J}}\dot{\theta}^2$$

$\sin^2+\cos^2=1$

$\bar{J}$ Generalized Inertia respect θ

{check dimensional consistency}

$[kg\,m^2]\,1/s^2 = [J]$

dissipation

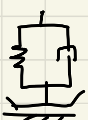
$$D = \frac{1}{2}c\,\underbrace{\Delta\dot{l}^2}_{} = \frac{1}{2}c\,4L^2\cos^2\theta\,\dot{\theta}^2$$

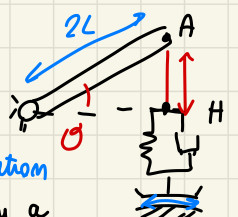
elongation of dumper

$$\Delta l = 2L\sin\theta + \Delta l_0$$

PRE-LOAD

both spring and dumper (parallel)





from $\theta = 0$ equilibrium + Generic θ Rotation

BUT SPRING compressed @ eq of Δl_0

cause using a SLIDER, JUST Vertical elongation

moves horizontally

total elong.
consider also the Δl_0 NOT depending on θ

$$\Delta l_0 = \text{constant} \rightarrow \Delta\dot{l} = 2L\cos\theta\,\dot{\theta}$$

potential en

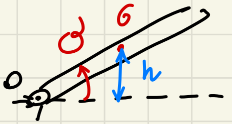
$$V = V_{el} + V_g = \frac{1}{2} k \Delta l^2 + mgh =$$

due to SPRING, elastic device

{ height.. vertical position of body center of mass }

h in general use a specific sign convention Respect fixed Reference → usefull to use the hinge!

height of G as GK



$h>0$ IF G above reference!

So if G above ref. the Rigid body have more V_g than in key horiz

IT depends on ϑ value ⇐ IF instead G below reference, we apply an $h<0$ distance

$$\begin{cases} \vartheta>0 : h>0 \\ \vartheta<0 : h<0 \end{cases} \qquad h = +L \sin(\vartheta)$$

$\vartheta>0$ G above point O reference

IF $\vartheta<0$ than G below and consistent to $\sin<0$ ($h<0$)

↓

$$V = \frac{1}{2} k (\Delta l_0 + 2L \sin\vartheta)^2 + mgL\sin(\vartheta) =$$

$$= \cancel{\frac{1}{2} k \Delta l_0^2} + 2L\Delta l_0 \sin(\vartheta) + \frac{1}{2} k 4 L^2 \sin^2(\vartheta) + mgL\sin(\vartheta)$$

$\frac{\partial}{\partial\vartheta} = 0$ CONSTANT

how to find Δl_0 PRELOAD (STATIC elongation of the spring so @ $\vartheta = 0$ equilibrium)

we can evaluate it in 2 ways... USING the THEOREM of

IN equilibrium

1. position, total potential energy is stationary! STATIONARY $\left.\frac{\partial V}{\partial \vartheta}\right|_{\vartheta=0} = 0$

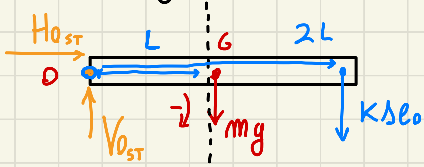
$$\left.\frac{\partial V}{\partial \vartheta}\right|_0 = k\Delta l_0 2\cancel{L} + \cancel{\frac{1}{2} k 4 L^2 \, 2 \sin\vartheta \cos\vartheta} + mg\cancel{L} = 0$$

$$\Rightarrow \boxed{\Delta l_0 = -\frac{1}{2}\frac{mg}{k}}$$

STATIC spring elong. to satisfy $\vartheta = 0$ equilibrium

STATIONARITY of V in eq position

② Related on how forces act on the system — EQUILIBRIUM @ eq state
drawing it @ equilibrium and drawing the static forces



+ POSIT. Moment

sign convention Δe + (positive if elongated)

representing spring elongated, $\Delta\ell_o > 0$ try to pull down the BAR

static condition...
NOT INERTIA, VISCOUS forces,
BUT don't forget that also
on equilibrium we have CONSTRAIN forces (STATIC)

MUST satisfy equilibrium
$$\begin{cases} \sum \vec{F} = 0 \\ \sum \vec{M}_c = 0 \end{cases}$$
↑ Better choice

find relationship between F_{el} and F_p → NOT use the CONSTRAINS! NOT useful to find

⇓

depends on our smartness to find the CORRECT terms

Using $\sum \vec{M}_c = 0$ with $C \equiv O$ center
IN FACT SMART choice

$M_c = \vec{F} \cdot \vec{b}$ with $\vec{b}$ = arm of the force, that if passing through point -- 0 moment

H, V does NOT contribute on the moment! → eliminated easily

$$\sum M_O = 0 \rightarrow -mg\cancel{L} - K\Delta\ell_o 2\cancel{L} = 0$$ NO CONSTRAIN contribution :) !

effect of ROTATING opposite to our convention (+)

$$\Rightarrow \boxed{\Delta\ell_o = -\frac{1}{2}\frac{mg}{K}}$$ as before

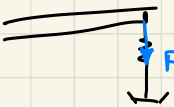
this ② advantages is to get a better phisical understanding of the result!

- showing that @ eq $\Delta\ell_o < 0$, SPRING compressed to carry bar Weight and the needed compression
- amount of compression @ eq such that $|K\Delta\ell_o| = \frac{1}{2}mg$ half the bar weight

($K\Delta\ell_o$ ↓ for convention, $\Delta\ell_o > 0$ ↑ so the force act from up to below) half weight because center of mass in between spring and hinge, LEVER Rule → so share the weight by hinge and spring

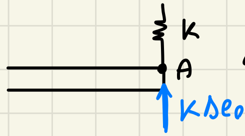
(center of mass in the middle) → carry mg by SPRING and HINGE

$\Delta l_0 > 0$ elongates



if $\uparrow K\Delta l_0$ means spring below A, issue say if stretch it moves up

IF INSTEAD

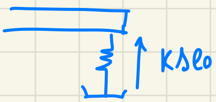


spring above A
$\Delta l_0 > 0$
push up point A

|| CAREFUL on sign convention || Δl_0 when in traction

IF instead setting → —WW— ←
+

positive if compressed so the drawing of Fel convention changes



but change also $\Delta l = -2L\sin\theta + \Delta l_0$

$\theta > 0$ means $\Delta l < 0$!

use always same choice! ⇒ usually best choice is $\Delta l_0 > 0$ IF elongating the spring

⇒ STATICAL elongation → FORCE pointing downward

we find $\Delta l_0 < 0$ to express how the Force is in opposite direction than chosen one because $mgL + K\Delta\ell_0 2L = 0$ (1) or (2) result

so in $V = \cancel{2L\Delta\ell_0 \sin(\theta)} + \frac{1}{2}K4L^2\sin^2(\theta) + \cancel{mgL\sin(\theta)}$

so we obtain $V = \frac{1}{2}K4L^2\sin^2(\theta)$ Total potential energy

WORK $\delta W_{nc} = 0$ No external t-dep. force acting

{LAGRANGE} all energetic quantities found

$$(mL^2+J)\ddot{\theta} + \emptyset + cL4L^2\underbrace{\cos^2\theta}_{\simeq 1}\dot{\theta} + K4L^2\underbrace{\sin(\theta)}_{\simeq\theta}\underbrace{\cos(\theta)}_{\simeq 1} = 0$$

system equation of motion (NON LINEAR)

→ LINEARIZE around $\theta = 0$ replacing each term with its power series expansion troncated @ 1st order therm!

$$\overbrace{(mL^2+J)}^{\bar{J}}\ddot{\theta} + \underbrace{4cL^2}_{\bar{c}}\dot{\theta} + \underbrace{4KL^2}_{\bar{k}}\theta = 0$$

LINEARIZED equation of motion in $\ddot{\theta}, \dot{\theta}, \theta$

yet linear

generalized parameter of system

$$\begin{cases} \cos(\theta)\big|_{\theta=0} \simeq 1 & \{\text{power series expansion of } \cos(\theta) \quad 1 - \cancel{\frac{\theta^2}{2!}} + \cancel{\frac{\theta^4}{4!}} \dots\} \\ \sin(\theta)\big|_{\theta=0} \simeq \theta \end{cases}$$

$\theta = 0$ valuated

⇒ SOLUTION of that equation describe $\omega = \sqrt{\dfrac{4KL^2}{mL^2+J}}$

undamp NATURAL FREQ

$\omega = \sqrt{\dfrac{\bar{K}}{\bar{J}}}$

APPROACH 2 || shortcut || (as seen in previous Theory lesson, how to directly linearize)

linearize eq of motion smart way → using CORRECT assumption directly when WRITING the energetic quantity

$$T = \frac{1}{2} m |\bar{V}_G|^2 + \frac{1}{2} J \dot{\vartheta}^2$$

We know that, looking the linearized eq of motion of the system

all linear in $\vartheta, \dot{\vartheta}$

we want find all linearized around $\vartheta = 0$ equilibrium (horizontal bar)

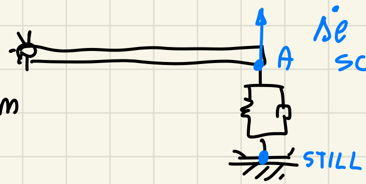
we wanna describe $\vec{V}_G(\dot{\vartheta})$ ← for RIVAL's theorem $\vec{V}_G \perp OG$ with magnitude $|\vec{V}_G| = L\dot{\vartheta}$

$$\Rightarrow \quad T = \frac{1}{2} m L^2 \dot{\vartheta}^2 + \frac{1}{2} J \dot{\vartheta}^2 = \frac{1}{2}(mL^2 + J)\dot{\vartheta}^2$$

$$D = \frac{1}{2} c \, \dot{\Delta\ell}^2 = \frac{1}{2} c \, 4L^2 \dot{\vartheta}^2$$

don't worry on considering generic position to evaluate $\dot{\Delta\ell}$ ⇒ we want to find JUST $\dot{\Delta\ell}(\dot{\vartheta})$ in special position (horiz.)

$\vec{V}_A = 2L\dot{\vartheta}$ for RIVAL's theorem



$\dot{\Delta\ell}$ is JUST the scalar velocity of A

$V = \dots$

both h G of G and elongation $\Delta\ell$ are related non linearly to ϑ with $\sin\vartheta$

$h = L\sin\vartheta$

$\Delta\ell = 2L\sin\vartheta$

involves $\sin(\vartheta)$

↳ because $\sin(\vartheta)$ such that $\left.\frac{\partial^2 \sin(\vartheta)}{\partial \vartheta^2}\right|_{\vartheta=0} = 0$

h, $\Delta \ell$ involves $\sin(\theta)$ → recause it 2ND derivative is 0 evaluated @ 0...
WE treat it with hessian function
defing $K_{el_{II}}$, K_g down to 0!

→ NO II order Kel or g elast. stifness ⇒ like NO weight
OR STATIC elongation

↓ JUST take K_{el_I}, linearized relationship

$$\Delta \ell = 2L\theta$$

$\sin\theta \simeq \theta$ I ORDER

NEGLECTING $\Delta \ell_0$ related to II ORDER
companion to weight!

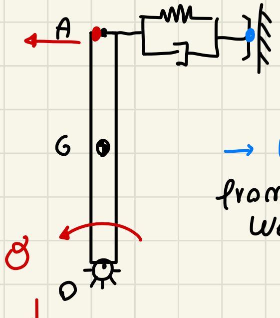
⇓

so $V = \frac{1}{2} K \Delta \ell^2 = \frac{1}{2} K 4L^2 \theta^2$

LAGRANGE equation (on this Linearized quantities)
↳ use shortcut
is very usefull to find rapidly our eq of motion!

es 2

fixed point on x displacement (moove along y)



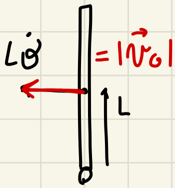
same system BUT rotaty by 90°
(equilibrium position in vertical displacement of har)
different equation of motion for weight force

→ use SHORT CUT

from mom linear kinematic relationship
We expect NO $K_{el\,II}$ because depends on $\sin(\theta)$
and ALSO we expect $K_{el\,II} = 0$ because
$\Delta l_0 = 0$ to have equilibrium at $\theta = 90°$
instead if $\Delta l_0 \neq 0$ it cause a moment
so loose equilibrium

↓
Reconsider IF there is gravitation stiffness!

$$T = \frac{1}{2} m |\vec{V}_G|^2 + \frac{1}{2} J \dot{\theta}^2 = \frac{1}{2}(mL^2 + J)\dot{\theta}^2$$

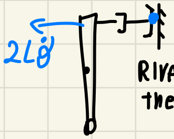


RIVALS Theorem

as before! same expression of es1 system

$$D = \frac{1}{2} c \Delta\dot{l}^2 = \frac{1}{2} c\, 4 L^2 \dot{\theta}^2$$

as Before short cut, NO change respect T, D in this syst



RIVALS theorem

positive elongation speed, increase dumper lenght

$\Delta\dot{l} = +2L\dot{\theta}$ because IT increases Δl

$$V = V_{el} + V_g$$

↓

$$V_{el} = \frac{1}{2} k \Delta l^2 = \frac{1}{2} k\, 4 L^2 \theta^2$$

(same as before!) $V_{el} = V_{TOT}$ of es1

⇓

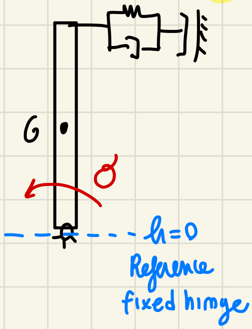
BUT in this system es2 respect the previous one we have also V_g contributes

$\Delta l_0 = 0$!

$$\Delta l = \int \Delta\dot{l} = 2L\theta + \cancel{const} \; 0$$

$\Delta l_0 = 0$
Integration constant

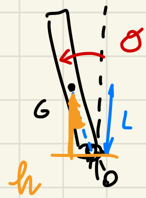
NOW we have V_g effect, because



With NOW ROTATED system V_g changes. SO we have NOT $\cancel{K_g=0}$

$$V_g = mgh = mgL\cos(\theta)$$

IN a general displacement (NO Linearz.)



(+) because $\theta>0$ means still $h>0$ and $\theta \geq \frac{\pi}{2} \rightarrow h<0$ coherent with OUR choice

$$h = +L\cos(\theta)$$

replace non linear expression by POWER SERIES approx

troncated NOT in I therm → but on II ORD therm when we bring V on Lagrange ... $\frac{\partial V}{\partial \theta} = \dots$

Because $(V_g + V_{el}) = V_{TOT}$ Take FIRST derivative, so to have first order approx (linearz) We need to take until II ORD, so derivate and lead to I ORD

$$\cos(\theta) \simeq \left(1-\frac{\theta^2}{2}\right)$$

$$V_g = mgL\left(1-\frac{\theta^2}{2}\right) = \underset{\partial/\partial\theta}{\overset{\text{const}}{\cancel{mgL}}} - \frac{1}{2}mgL\theta^2$$

Lagrange:

$$\underbrace{\frac{d}{dt}\left(\frac{\partial T}{\partial \dot\theta}\right) - \cancel{\frac{\partial T}{\partial \theta}}} + \frac{\partial D}{\partial \dot\theta} + \frac{\partial V}{\partial \theta} = \cancel{\frac{\delta W_{NC}}{\delta \theta}} = 0$$

Non time dep forces

(eq of motion)

$$(mL^2+J)\ddot\theta + 4L^2c\dot\theta + 4L^2k\theta - mgL\theta = 0$$

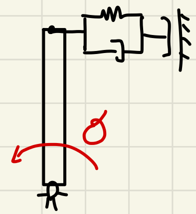
New term respect es 1

check for dimensional coherence!

$$(mL^2+J)\ddot\theta + 4L^2c\dot\theta + (4L^2k - mgL)\theta = 0$$

$$(mL^2+J)\ddot{\theta} + 4L^2c\,\dot{\theta} + (4L^2k - mgL)\theta = 0$$

different GRAVITY ROLE

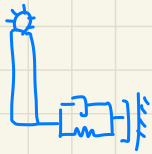


INVERSE PENDULUM → $K>0$ created by spring $K_{el}>0$

$K_g<0$ destabilized spring

↑ decrease STIFF

IF instead



$+K_g$ act as positive STIFNESS

the weight increase STIFF

⇓

$$\omega = \sqrt{\frac{4L^2k - mgL}{mL^2+J}}$$

← $\omega_{new} < \omega_{est}$ previous case

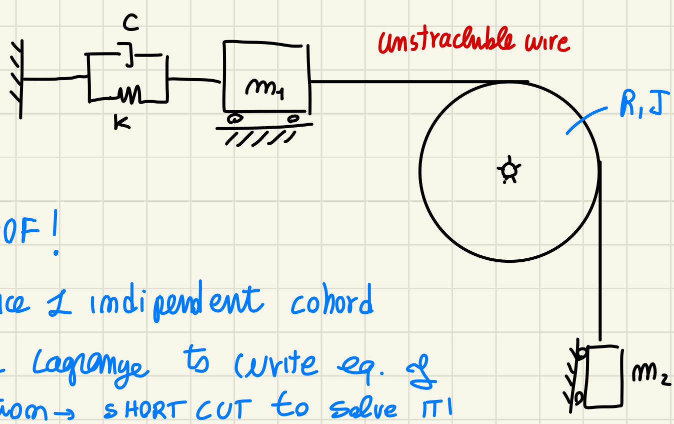
New ω depends on Relative new values

IT can happen $mgL > 4L^2k \rightarrow K_{TOT}<0$ causing INSTABILITY of the system!

We expect $K_{TOT}>0$ to have a STABLE SYSTEM!

⇓

In a New 1DOF sistem like



1 DOF!

Choice 1 indipendent cohord

Use Lagrange to write eq. of motion → SHORT CUT to solve IT!